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Empirical Analysis of Classical and Quantum Algorithms for Portfolio Optimization: Enhancing Financial Decision-Making through Quantum Computing

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***Abstract*—This research investigates the efficacy of quantum and classical algorithms in the context of portfolio optimization, focusing on a dataset comprising 20 equities from India's National Stock Exchange over a duration of three years. The study employs quantum methodologies such as the Quantum Approximate Optimization Algorithm (QAOA), the Variational Quantum Eigensolver (VQE), and the Exact Minimum Eigen Solver, alongside conventional strategies including Mean-Variance Optimization, Genetic Algorithms, and Simulated Annealing. The methodology for this study includes data preprocessing, issue designing, and performance evaluation based on metrics such as MSE, RMSE, and Mean Absolute Error. The results obtained show quantum algorithms exhibit better performance over the classical methods regarding prediction accuracy and risk-return profiles, especially the Exact Minimum Eigen hSolver and QAOA. The Exact Minimum Eigen Solver and QAOA had the lowest MSE values (0.006 and 0.005, respectively), whereas the best-performing classical technique, the Genetic Algorithm, had an MSE of 0.0105. These findings imply that quantum algorithms provide significant advantages in portfolio optimization problems, with the potential to revolutionize financial optimization tactics. The research can be further unfolded into avenues for future inquiry, including expansion of the choice of assets, alternative data sources integration, as well as hybrid quantum-classical models to efficiently enhance portfolio management strategies.**

***Index Terms*—Quantum Algorithms, Portfolio Optimization, Classical Approach**

I. Introduction

Quantum algorithms function based on the principles of quantum mechanics to process information in ways impossible in classical algorithms. Some types of quantum algorithms utilize quantum bits or qubits. A qubit is comparable to the classical bit used in a computer except that, by its property called superposition, it can exist in multiple states at any given time. Qubits also have the property of entanglement, allowing quantum systems to perform complex calculations exponentially faster than any existing computer. Quantum algorithms are under research for possible applications within evolving industries that demand high processing such as encryption, drug discovery and especially finance.

The origin of quantum algorithms dates back to the early 1980s, when physicist Richard Feynman proposed that quantum systems can only be simulated in practice by other quantum computers. This idea laid down the basis for quantum computing as a field of research. In 1994, Peter Shor designed the first quantum algorithm with practical application-the Shor's algorithm that can factor huge integers tenfold faster than the best known techniques in classical computation. This discovery proved vital, as researchers needed to explore more complex computer-related issues, including optimization, machine learning, and cryptography, using quantum algorithms.

Quantum algorithms are the novel application in finance, though rising in popularity. Two major quantum algorithms currently researched for optimizing portfolio management, risk assessment, and derivative pricing include the QAOA and VQE. These algorithms exploit parallelism and entanglement-properties characteristic of quantum systems-for solutions unattainable by classical computers.

The finance sector actively researches quantum algorithms for solving problems requiring huge amounts of data or complex computation. Portfolio optimization is an important task in finance, involving choosing the optimum portfolio of options which maximizes return but minimizes risk. Traditional methods, including the Markowitz model, rely on classical algorithms which are computationally constrained in

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large datasets or with complex constraints.

Classical algorithms have been the foundation of financial computations for decades. The most general class of classical portfolio optimization techniques includes linear programming, dynamic programming and heuristic methods. The Markowitz Mean-Variance Optimization, published in 1952, employs quadratic programming to reduce portfolio variance for any desired expected return. With more assets, complexity increases exponentially, creating an inability to solve optimal portfolio optimization problems.

Quantum algorithms can surpass traditional algorithms in portfolio optimization. Quantum parallelism allows examination of many alternatives simultaneously, considerably reducing the time to obtain the optimal portfolio. Algorithms such as QAOA are designed specifically to tackle combinatorial optimization problems, making them suitable for portfolio selection and asset allocation.

Recent years have shown increased interest in portfolio management using artificial intelligence, especially ML. ML algorithms process vast amounts of data, recognize patterns and trends, and make predictions on market movements, enhancing risk assessment and portfolio optimization for better financial decision-making. As AI analyzes big data in real time, portfolio management becomes more dynamic, allowing asset managers to respond timely to market changes.

This paper presents a comparison of classical and quantum methods to optimize portfolios. The research involves developing a quantum algorithm based on QAOA and comparing it with traditional algorithms, namely Mean-Variance Optimization and Genetic Algorithms. The study focuses on computing time, optimum accuracy for portfolios, and scaling. The methodology includes analyzing historical financial data using quantum and classical algorithms for portfolio optimization, followed by performance evaluation to determine potential advantages in solution speed and quality.

II. Literature Survey

The use of quantum algorithms in portfolio optimization has been extensively studied in recent literature, with several approaches being evaluated to improve the efficiency and effectiveness of financial decision-making [1], [2], [3], [4]. The PO-QA framework, which uses VQE and QAOA to optimize portfolio selection by adjusting quantum circuit parameters, shows QAOA outperforming VQE in efficiency [1]. The study establishes links between quantum circuit complexity and portfolio performance, while emphasizing the need for further research into quantum circuit designs and scalability.

Certo et al. [2] study quantum annealers for portfolio optimization, offering a QUBO model that considers real-world restrictions including fundamental analysis metrics and sector investment limits. The study reveals classical algorithms, particularly those using commercial optimization tools like CPLEX, outperform existing hybrid quantum methods in static portfolio optimization settings, highlighting the maturity of conventional methods.

Wu et al. [3] present a hybrid quantum-classical approach for computing VaR and CVaR for European option portfolios. This method addresses high-dimensionality difficulties in risk estimation by combining quantum algorithms with standard partial differential equation techniques, though its applicability to American options remains limited.

Owhadi-Kareshk and Boulanger [4] introduce PortFawn, an open-source Python toolkit for portfolio optimization utilizing both classical and quantum computing techniques. The study demonstrates quantum annealing application in portfolio optimization using a QUBO model, noting the focus on theoretical rather than practical performance in existing literature.

A novel quantum method for portfolio management shows improved runtime complexity compared to classical algorithms [5], though implementation challenges persist regarding quantum hardware robustness and input matrix conditioning. Wilkens and Moorhouse [6] explore quantum computing applications in financial risk measurement using quantum amplitude estimation and Monte Carlo simulations, identifying current technological limitations in scalability and reliability.

Carrascal et al. [7] demonstrate quantum algorithms matching or slightly surpassing classical methods, particularly with increasing asset numbers, while acknowledging current hardware limitations. Shunza et al. [8] present the QMOA for discrete portfolio optimization, showing effective asset selection through a network-based approach, though questions remain about scalability with larger asset pools.

Duarte Frazão [9] examines QAOA and QAOA+ for investment portfolio optimization, demonstrating QAOA+'s superior performance in execution time and solution quality. A systematic review [10] emphasizes QAOA and HHL applications while highlighting challenges such as decoherence and negative eigenvalue handling in matrices, underscoring the need for refined quantum algorithms and improved practical implementation in financial decision-making.

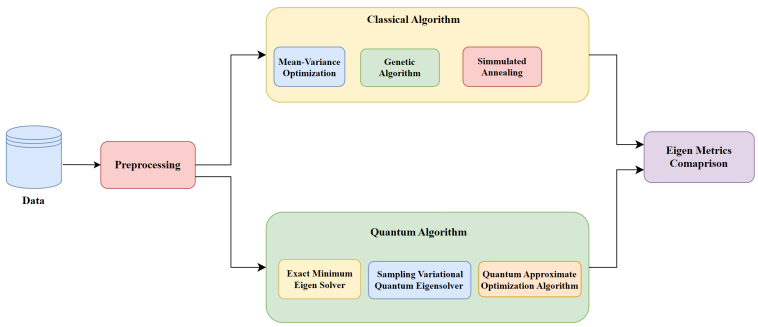

Fig. 1. Proposed approach for analysis of Quantum and Classical Algorithms for Portfolio Optimization

III. Proposed Approach

Figure 1 describes an outline of the methodology for the proposed approach to portfolio optimization based on quantum and classical algorithms. The methodology compared the performance of some quantum algorithms, including Quantum

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Approximate Optimization Algorithm and Variational Quantum Eigensolver, to the related work that relied on classical algorithms, such as Mean-Variance Optimization and Genetic Algorithms, to compare their relative performance on a simulated dataset of asset returns. Sharpe Ratio, Sortino Ratio, and maximum drawdown are the measurements adopted to assess which of the techniques is more capable.

A. Data Collection and Preprocessing

Historical data was generated using adjusted closing prices for 20 selected stocks from the National Stock Exchange of India (NSE) spanning January 1, 2021 to January 1, 2024, obtained through the yfinance library. The selected tickers include: ADANIENT.NS, BAJFINANCE.NS, HCLTECH.NS, HDFCBANK.NS, HDFCLIFE.NS, HEROMOTOCO.NS, HINDALCO.NS, HINDUNILVR.NS, ICICIBANK.NS, INDUSINDBK.NS, POWERGRID.NS, RELIANCE.NS, SBIN.NS, SUNPHARMA.NS, TCS.NS, TATACONSUM.NS, TATAMOTORS.NS, TATASTEEL.NS, TECHM.NS, WIPRO.NS. Data completeness was verified, and missing values were identified and excluded from analysis. Daily returns were calculated as percentage changes in adjusted closing prices to understand asset volatility and risk, providing a foundation for mean-variance optimization. The preprocessing phase included normalization and standardization of return data to ensure appropriate scaling, crucial for both quantum and classical optimization to prevent asset dominance based on size.

B. Setting Up the Optimization Problem

Calculating the covariance matrix of daily returns and predicted returns for each asset yields the optimization issue. The covariance matrix offers information about how the assets move together, which is critical for analyzing portfolio risk. The risk factor (q) is set at 0.5 to balance expected return with portfolio risk. The portfolio's budget is set at half the number of assets, allowing for a more diverse allocation strategy without over-concentration on any particular asset.

C. Different Algorithms Used

The choice of the algorithms for this work was based on demonstrated effectiveness in the portfolio optimization. Among them, classical algorithms such as MVO, GA, and SA have been broadly used for financial optimization because they have shown robust performance within smaller data sets and within deterministic problem-solving capabilities. These quantum algorithms - QAOA, VQE, and Exact MES-were chosen for the task because they can look at a number of solutions simultaneously through quantum superposition, so they could potentially produce efficient and accurate results in problems involving complex portfolio optimization. Optimization process: the quantum and classical algorithms differ. A classical algorithm, such as MVO, can be optimized by minimizing portfolio risk through the covariance matrix maximized return according to the historical data. On the other hand, quantum algorithm, such as QAOA and VQE, encodes the quantum circuit of the optimization problem in the portfolio. Therefore, by iteratively adjusting the quantum gates allows the system to converge to the best solution. This hybrid approach, combining classical data preprocessing with quantum optimization, creates the possibility of exploring several solutions in parallel and therefore contributes to an increase in the ability of complex and large datasets by the algorithm.

1) Quantum Algorithms:

1) **Exact Minimum Eigen Solver:**
This algorithm utilizes the NumPyMinimumEigensolver to find the optimal portfolio allocation. The mathematical representation of the objective function can be expressed as:

$$\text{Maximize } f(x) = \mu^T x - q \cdot x^T \Sigma x \tag{1}$$

2) **Sampling VQE:**
The hybrid quantum-classical algorithm, called the Variational Quantum Eigensolver (VQE), is designed to find approximate solutions to challenging optimization problems by minimizing a quantum system's energy. In the course of VQE, a classical optimizer iteratively changes the parameters of a quantum circuit in order to minimize an objective function that typically represents the portfolio's variance or risk. This hybrid strategy makes purely classical methods much less effective in exploring the space of solutions.

3) **QAOA:**
The Quantum Approximate Optimization Algorithm focuses on solving combinatorial optimization issues through a unique approach. This approach works by creating a quantum state that encodes the answer to the optimization problem and then iteratively optimizing the circuit to enhance the objective function.

While both VQE and QAOA are quantum algorithms designed for optimization problems, they differ in their applications. VQE is more suited to continuous optimization problems, as it minimizes the energy of a quantum system using a hybrid quantum-classical approach. In contrast, QAOA is specifically designed for combinatorial optimization problems, where the goal is to find the optimal discrete solution among a set of possibilities. Both algorithms exploit quantum mechanical properties to enhance the efficiency of the optimization process, though they are applied in different financial and computational contexts.

2) Classical Algorithms:

1) **Mean-Variance Optimization (MVO):**
The MVO approach focuses on maximizing the expected

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return for a given level of risk. The optimization problem is represented as:

$$\text{Maximize } f(w) = \sum_{i=1}^{n} w_i \mu_i - q \cdot w^T \Sigma w \qquad (2)$$

subject to the constraint $\sum w_i =$ Budget. Here, the objective is to find the optimal weights w for the portfolio that balance risk and return effectively, adhering to the principles laid out by Harry Markowitz.

2) **Genetic Algorithm (GA):**
This algorithm takes a population-based approach, imitating natural selection to evolve portfolio weights across generations. The fitness of each candidate portfolio is assessed based on its predicted return and risk, and the fittest portfolios are chosen for replication. This approach is especially beneficial for navigating complex, nonlinear optimization environments.

3) **Simulated Annealing (SA):**
Simulated Annealing is a probabilistic optimization technique that simulates metallurgical annealing. It explores the weight space by accepting both increases in portfolio performance and some deteriorations, allowing the algorithm to avoid local minima. The acceptance probability lowers over time, similar to cooling in physical systems, resulting in more refined solutions.

D. Evaluation Metrics and Performance

The performance of each algorithm is assessed using the expected future returns for each portfolio. The key metrics include:

1) **Mean Squared Error (MSE):** Calculates the average square of the errors between projected and actual results.
2) **Root Mean Squared Error (RMSE):** This metric measures the average error in the same units as the predicted values, making it easier to interpret.
3) **Mean Absolute Error (MAE):** Represents the average magnitude of the errors without regard to their direction, providing a straightforward interpretation of forecast accuracy.

These measures will be used to compare expected returns to a benchmark (an equal-weight portfolio) and determine the efficacy of each optimization technique. The comparison provides insights into the relative benefits and performance capabilities of quantum versus classical techniques in portfolio optimization.

E. Results

1) Optimal Portfolio Weights: Here the ideal portfolio weights determined by each of the six algorithms. The figures below show the asset allocations determined by each technique. These allocations represent the ideal capital distribution across the selected assets, as determined by the algorithms' risk minimization or return maximization.

TABLE I
PERFORMANCE EVALUATION OF ALGORITHMS USING MSE, RMSE, AND MAE METRICS

Algorithms	MSE	RMSE	MAE
Exact MES	0.006	0.245	0.196
Sampling VQE	0.007	0.264	0.211
QAOA	0.005	0.224	0.179
MVO	0.946	0.973	0.778
GA	0.0105	0.1025	0.082
SA	0.65	0.806	0.645

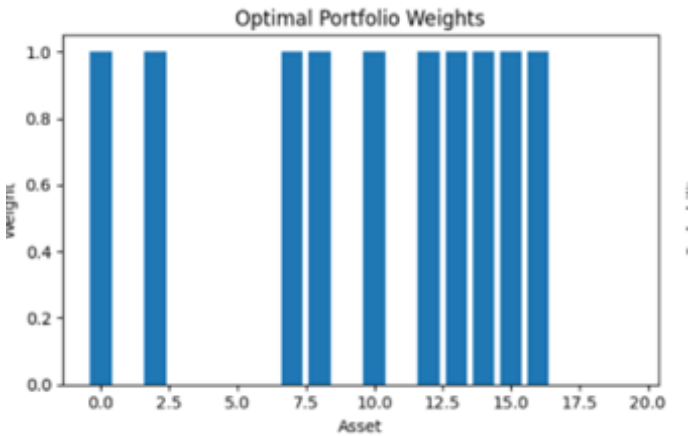

Fig. 2. Exact Minimum Eigen Solver

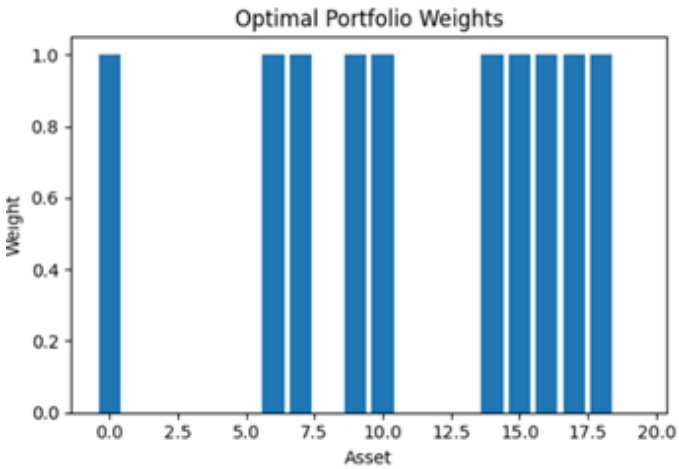

Fig. 3. Sampling VQE

2) Portfolio Return Comparison: The portfolio returns achieved by each of the six optimization strategies to determine their effectiveness in maximizing returns while controlling risk were studied. The figure below displays the variations in returns based on the portfolio compositions chosen by each technique, providing a visual depiction of their relative performance. By analyzing these returns, it can be determined how well each algorithm balances risk and reward, which is critical for effective portfolio management. This comparison

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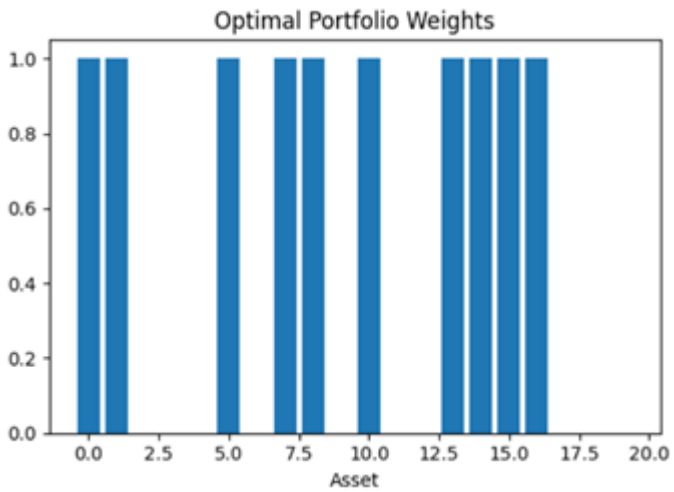

Fig. 4. QAOA

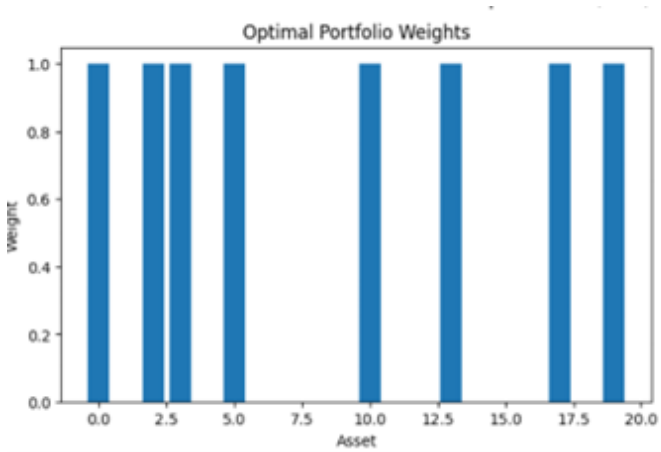

Fig. 5. MVO

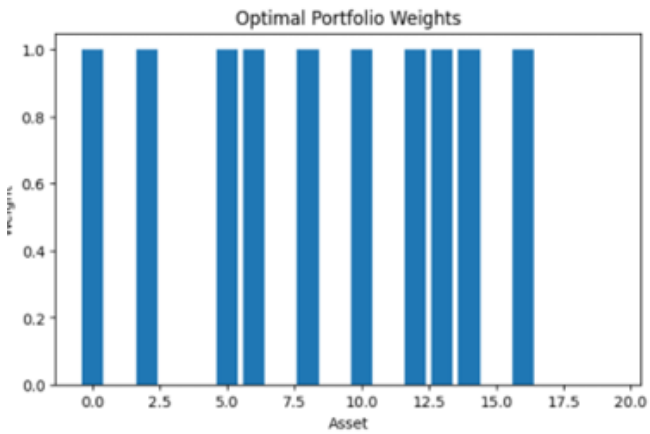

Fig. 6. GA

also demonstrates how each strategy responds to changing market conditions, revealing their robustness and capacity to adjust to various risk-return trade-offs.

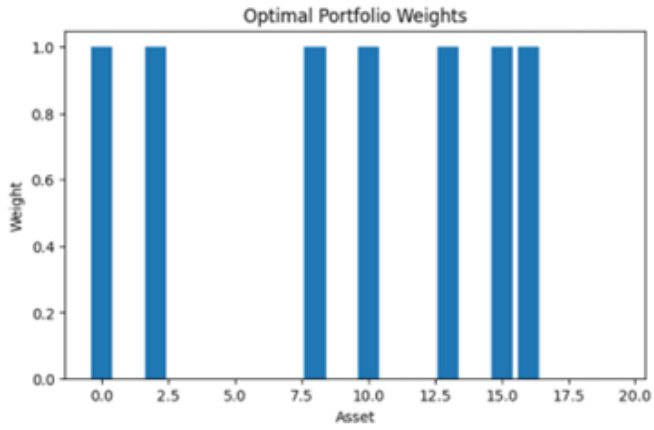

Fig. 7. SA

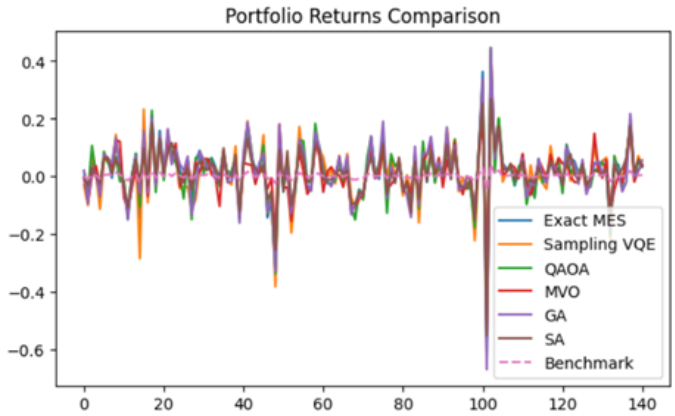

Fig. 8. Returns Comparison

3) Error Metrics Comparison: The error measures (MSE, RMSE, and MAE) for each method were compared. The graphic below depicts these error levels, allowing for a clear comparison of each method's precision and accuracy while optimizing portfolio weights. Lower error measures indicate greater performance, demonstrating how closely each algorithm's output matches the optimum portfolio solution. Quantum algorithms gain great precision purely out of the fact that it can scan an enormous solution space parallelly. There are quantum algorithms, for example, QAOA and VQE: these update the parameters of a quantum circuit iteratively so that the quantum circuit minimizes the objective function of the portfolio optimization problem, typically variance or risk. These algorithms apply quantum superposition and entanglement, enabling one to evaluate many candidate solutions, which makes them more accurate compared to the classical algorithms in complex and highly dimensional data sets.

The two types of methods also differ from each other in terms of the reliability of the optimization process. Classical algorithms, such as MVO and GA, are well proven to be deterministic and give constant results when used on small and medium-sized datasets. On the contrary, quantum algorithms

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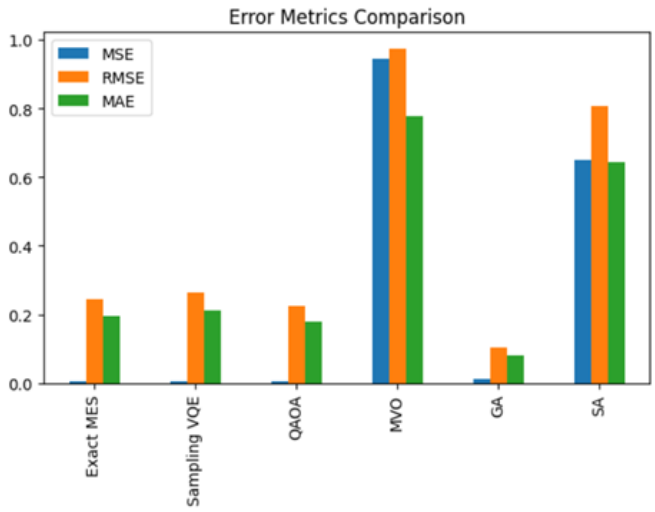

Fig. 9. Error Metrics

may look new, but they still face problems like quantum noise and are becoming reliable with advancements in hardware. Hybrid models where classical algorithms first perform preprocessing of data and then quantum algorithms for optimization make such methods highly reliable and scalable, especially when applied to large and complex datasets.

IV. Conclusion

This study explored the comparative effectiveness of quantum and traditional algorithms in portfolio optimization through suitable allocations. Intensive research and evaluation through various measures identified different performance characteristics among the methodologies. MES and QAOA demonstrated quantum supremacy in predictive accuracy, outperforming other algorithms with MSE values of 0.006 and 0.005 respectively. These strategies produced better balanced risk-return profiles. The Sampling Variational Quantum Eigensolver showed comparable but lower accuracy with 0.007 MSE. Among traditional approaches, the Genetic Algorithm recorded the highest MSE at 0.0105, indicating challenges in hard optimization landscapes and lower precision compared to quantum counterparts. Mean-Variance Optimization (MVO) and Simulated Annealing (SA) yielded significantly higher error metrics, demonstrating the relative inefficiency of traditional approaches. The research validates that quantum algorithms can notably enhance portfolio optimization operations through reduced prediction errors and improved performance metrics. The results indicate quantum computing's potential to transform financial optimization strategies.

Future Scope

Although this research establishes a framework for understanding quantum algorithms' potential in portfolio optimization, further investigation is required. Future research directions include expanding the asset range beyond the current 20 selected tickers to equities across multiple industries, enhancing the scope of asset selection analysis. Incorporating alternative data sources, such as social media sentiment and macroeconomic indicators, could improve predictive power and provide comprehensive insight into asset return drivers.

Exploration of hybrid models combining strengths of quantum and classical approaches merits investigation. Classical approaches could be applied for preliminary optimization, followed by quantum technique refinement. Further advancement in quantum computing technology remains essential for testing these algorithms in real trading scenarios to establish scalability and live performance. As advanced quantum techniques evolve, future research could leverage new developments to enhance portfolio management approaches.

Sensitivity testing to verify optimal portfolios' response to market conditions or asset correlation changes would establish solution credibility. Addressing these aspects would advance the application of quantum algorithms in finance and establish foundations for efficient portfolio management solutions.

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